

MicroPulse — Indoor Load Guide

Reading the Football Movement Profile-driven indoor load page

MicroPulse Performance Intelligence Platform

April 2026

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1. What this page tells you

The Indoor Load page (Monitoring → Indoor Load) exists because **GPS does not work indoors**. When your team trains in a sports hall, the pods cannot resolve velocity bands, sprint distance or any of the GPS-based load metrics the rest of the platform leans on.

That doesn't mean the players aren't loading themselves — they are. It just means the *signal* has to come from a different place. This page reads load entirely from the IMU (the inertial sensor inside every Catapult pod), grouped by **Football Movement Profile (FMP)** — Catapult's classification of what the player was doing at every moment of the session: standing, walking, running at moderate pace, running hard, dynamic short bursts at low / medium / high intensity, etc.

It then combines those FMP categories into a single **Composite Indoor Load Score** that is comparable across days, sessions, and seasons. Same player on the same drill should produce the same score whether the data came from grass or hall.

For an outdoor-only club this page may not appear in your menu at all. For Icelandic clubs and any northern-Europe winter season club, this page becomes the most-used screen on the platform from October to April.

2. The Composite Indoor Load Score

The composite is a weighted sum of:

- **Player Load** (raw IMU accelerometer signal)
- **Dynamic-High % time** (% of session spent in the FMP “dynamic high” band — short bursts at high intensity)
- **HMLD** (High Metabolic Load Distance, when the IMU can compute it)

- **IMA Free Running Bands 7+8** (high-intensity stride detection — used as the indoor “sprint count” proxy)
- **IMA Decel Bands 2+3** (medium and high-intensity decelerations)

The weights were tuned in late 2024 against a reference dataset of paired indoor-outdoor sessions to make the composite score *interchangeable* between training environments. A player with composite=85 on grass and composite=85 in the hall did the same amount of work to their body, even though the underlying GPS signals differ entirely.

This means everything downstream — ACWR, personal baselines, McBurnie, decision verdict — works seamlessly across indoor/outdoor weeks. You do not have to think about which environment the data came from.

3. The page top-to-bottom

3.1 Team Status Banner

A single colored band at the top with a one-sentence summary:

“15 ready, 2 lighter, 1 rest, 2 injured, 0 sick”

Color is driven by team severity — emerald (everything fine), amber (some concerns), rose (multiple players need protection).

Pill badges next to the count make the readiness flag distribution visible at a glance.

3.2 Top Concerns

A 3–5 player list with the single most important reason each is flagged. Examples:

“Jónatan — Spike (æfði miklu meira) — composite +1.8 SD”

“Bjarki — Heavy + elevated ACWR (1.43)”

“Aron — Light (under-stimulated, 4 sessions in 14d)”

Each entry links to the player’s expanded row below.

3.3 Squad-level summary tiles

Five small tiles giving you the indoor data shape at a glance:

- **Heavy / Spike** — players whose composite is significantly above their personal baseline today
- **Typical** — players within personal range
- **Light / Below avg** — players doing less than their personal baseline
- **Indoor sessions (7d)** — count of indoor sessions in the last week, team-wide
- **Indoor (28d)** — count over the last 28 days

Use these as the team periodisation read. If “Heavy” jumps over 5 players in one day, the previous session was harder than planned.

3.4 Per-player rows

Each player gets a collapsible row. The header always shows:

- Player name
- Action label (△ **Righty** // Rest / Sick / Injured)
- Composite score badge

Click to expand and you see the full detail: composite vs personal baseline, latest session breakdown, 7-day cumulative, FMP movement distribution bars, indoor McBurnie proxy, ACWR.

3.5 14-day team heatmap

At the bottom of the page, a grid of cells: one row per player, one column per day. Each cell is colored by composite score:

- Light grey — no session that day
- Pale color — Light or Below
- Mid color — Typical
- Saturated — Heavy
- Deep saturated — Spike

A red inner border on a cell indicates an indoor session (vs the same color cell with no border, which would be an outdoor session at equivalent composite).

This is the single best view in the platform for spotting team-wide patterns: weeks where the squad over-trained, periods when too many sessions stacked, individual players who slipped under the radar with too many light days in a row.

4. Reading a player row in depth

When you expand a player row, you see five blocks:

4.1 “Ready today?” banner

A colored sentence, e.g. “*Ready for full training, baseline+12%.*” Color matches the action label.

4.2 Why

Plain-language drivers: “*Composite +0.9 SD vs baseline. ACWR 1.31 (within range). McBurnie A:D coupling YELLOW (0.58).*”

If you read nothing else, read this — it tells you exactly which signal moved.

4.3 Coach guidance

The “what to do” sentence: *“Full training OK. Drop the deceleration emphasis at the end of the session — coupling is borderline.”*

This is generated from the combined signals. Treat it as a starting point for discussion.

4.4 Latest session breakdown

A small grid showing the latest indoor session’s metrics:

Metric

Composite Indoor Load

Player Load

Duration (min)

Dynamic High %

HMLD (m)

IMA total

Decel Band 2+3 count

4.5 FMP movement distribution

A horizontal stacked bar showing how the player spent the session:

- **Very Low** — standing, walking
- **Low Intensity** — easy jog
- **Running Medium** — match-pace running
- **Running High** — high-speed running
- **Dynamic Low** — short bursts at low intensity (cuts, decelerations at submax)
- **Dynamic Medium** — short bursts at medium intensity
- **Dynamic High** — short bursts at high intensity (the most damaging single category)

A healthy hall session is roughly: 25% Very Low, 25% Low, 30% Running Med, 8% Running High, 7% Dynamic Low, 4% Dynamic Med, 1% Dynamic High. The *Dynamic High* % is the most informative single number — it is what separates a recovery session from a hard one.

4.6 Indoor McBurnie proxy

The McBurnie D:Sprint coupling reformulated for indoor:

Decels per minute / Dynamic-High minutes per minute

Same threshold logic as the outdoor McBurnie — values above ~2.5 indicate too many braking events per high-intensity stride, an early hamstring warning. The indoor version is

generally noisier than the outdoor version (IMU has lower resolution than GPS for sprint detection), so trust trend over single-session readings.

4.7 ACWR

Acute (7-day) divided by chronic (28-day) average. Same Gabbett 2017 framework as elsewhere on the platform; same thresholds:

- Below 1.3 — healthy adaptation
- 1.3–1.5 — watch list
- Above 1.5 — danger zone

For indoor data the ACWR is computed over indoor sessions only — it does not mix with outdoor data, because the load patterns are too different.

5. Workflow — how to use this in a normal indoor week

5.1 Daily — after each indoor session

Open the page within an hour of the session ending (Catapult sync usually completes that fast). Read:

1. The team status banner — overall read
2. Top Concerns — anyone the system flagged
3. The 14-day heatmap — pattern recognition

Total time: 2–3 minutes if everything is settled.

5.2 Weekly — Monday planning

Open the page at the start of every week before the planning meeting. Look for:

- Players whose 7-day session count is low (under-stimulation risk)
- Players whose ACWR is climbing across the heatmap
- Team-wide trend — is the cluster of “Heavy” cells growing?

This is where you set the week’s training intensity bias for indoor sessions.

5.3 Outdoor → indoor transition (autumn)

When you move from outdoor to indoor for the winter, give the system 2–3 sessions to recalibrate the personal baselines. The first indoor week may show “Insufficient data” for the indoor-specific personal baseline. By session 4 the personal baseline begins to stabilise.

You can speed this up if you have historical indoor data — talk to your performance support about backfilling.

6. Training mode selector — settings page

The pipeline auto-detects indoor vs outdoor for each session based on FMP signature and GPS quality. In 95% of cases this is correct. For the 5% where it isn't (e.g. an outdoor session in a stadium with poor GPS reception), you can override via:

Settings → Team training mode → set explicitly to "Indoor" or "Outdoor"

This override applies to the next session. After 24 hours it reverts to auto-detect.

7. Troubleshooting

“All my players show ‘Insufficient data’”

You don't have enough indoor sessions yet for personal baselines. The page needs ~14 indoor sessions per player to compute. Until you reach that, the page will populate progressively as more sessions accumulate. There's no manual workaround — just keep training and the system catches up.

“FMP movement distribution looks wrong (too much Very Low)”

Most often: the Catapult pod was on but the player was off the pitch for a long stretch (sub, injury timeout, etc.). The pod still recorded standing time and FMP counted it as Very Low. If the session metadata in OpenField correctly identifies the active period, the system uses that — otherwise it uses the full pod-on duration.

If a session is consistently miscategorised, talk to your performance support about adjusting the period boundaries in OpenField.

“Composite indoor load score doesn't match what I saw at training”

The composite is a *weighted* combination, not a single signal. A player who did a lot of distance but no high-intensity work will rate lower than one who did less distance but more dynamic-high. That mismatch is intentional — the composite is built to predict tissue load, not to count steps.

If you want to see the underlying signals, expand the player row. The latest-session table shows each component separately.

“Decel B2+3 count zero for everyone”

Two possible causes:

1. Your team has IMA accel/decel band reporting disabled in OpenField. Add the parameters and re-sync.
2. You're on older Catapult pods that don't compute IMA bands. The composite will still work — it just falls back to weighting Player Load + Dynamic High % more heavily.

“Heatmap shows red inner borders on outdoor sessions”

The inner border indicates *the system thinks it was indoor*. If you're seeing it on a session you know was outdoor, the auto-detect heuristic missed. Use the Settings → Team training mode override (above) to correct it on a per-session basis.

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